MOBILE SIGINT PLATFORM

Phased Implementation Plan

From Concept to Operational System

Version 1.0

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1. Executive Summary

This implementation plan provides a structured, phased approach to building the Mobile SIGINT Platform. The project is divided into five phases, each delivering incremental, testable functionality.

1.1 Phase Overview

Phase	Focus Area	Duration	Dependencies
1	Core Vehicle System	2-3 weeks	None
2	KrakenSDR Integration	1-2 weeks	Phase 1
3	ADS-B Station	1 week	Phase 1
4	Sensor Mesh Network	2-3 weeks	Phase 1
5	HaLow Mesh + Advanced	2-4 weeks	Phases 1-4

1.2 Total Estimated Timeline

- Minimum: 8 weeks (aggressive, parallel work on Phases 2-4)
- Typical: 12 weeks (sequential with integration time)
- Conservative: 16 weeks (including buffer for issues)

2. Phase 1: Core Vehicle System

PHASE 1 - Foundation for all subsequent work

2.1 Objectives

- Establish Main Controller (Pi 5) as the central hub
- Integrate BCD996P2 scanner with touch-friendly driving UI
- Set up Kismet for WiFi/BLE monitoring
- Configure GPS for position and timing
- Deploy to vehicle in Pelican case

2.2 Hardware Required

Item	Qty	Est. Cost	Source
Raspberry Pi 5 8GB	1	On hand	-
256GB NVMe SSD + HAT	1	$45	Amazon
7" Touch Display	1	$70	RPi Foundation
USB GPS (u-blox VK-162)	1	$18	Amazon
Netgear GS308 Switch	1	$30	Amazon
Pelican Case (1500)	1	$100	B&H
12V→5V 5A Converter	1	$15	Amazon
Cables, misc	-	$30	Various

Phase 1 Estimated Cost: ~$313

2.3 Implementation Steps

2.3.1 Week 1: Base System Setup

☐ Flash Raspberry Pi OS Lite (64-bit) to NVMe SSD
☐ Configure hostname: main-controller, static IP: 192.168.1.50
☐ Enable SSH with key-based auth
☐ Install and configure gpsd with USB GPS
☐ Install Kismet and configure for WiFi monitoring

2.3.2 Week 2: Scanner Integration

☐ Connect BCD996P2 via USB, identify serial port
☐ Create scanner-ui Flask application

☐ Implement hold/scan/lockout controls
☐ Configure auto-start and kiosk mode

2.3.3 Week 3: Vehicle Deployment

☐ Design and cut Pelican case foam inserts
☐ Wire 12V power distribution
☐ Install and test in vehicle
☐ Verify thermal performance

2.4 Validation Criteria

- Scanner UI displays frequency within 500ms of change
- System survives 2-hour driving test without issues
- System boots to operational state within 90 seconds

3. Phase 2: KrakenSDR Integration

PHASE 2 - Direction Finding Capability

3.1 Objectives

- Set up dedicated Pi 5 for KrakenSDR processing
- Configure KrakenSDR DOA software
- Integrate scanner-to-Kraken frequency handoff
- Display bearing on driving UI

3.2 Hardware Required

Item	Qty	Est. Cost	Source
Raspberry Pi 5 8GB	1	On hand	-
KrakenSDR 5-channel	1	$350	CrowdSupply
5x Matched dipole antennas	1 set	$100	KrakenRF
Magnetic antenna mount	1	$50	Various

Phase 2 Estimated Cost: ~$520

3.3 Implementation Steps

☐ Flash KrakenSDR image to Pi 5

☐ Configure hostname: kraken-host, IP: 192.168.1.51

☐ Connect KrakenSDR, run calibration

☐ Add Kraken tuning API to scanner-ui

☐ Implement one-button DF handoff

☐ Add bearing display to driving UI

3.4 Validation Criteria

- Kraken tunes to frequency within 500ms
- Bearing accuracy within 10 degrees on test signal

4. Phase 3: ADS-B Station

PHASE 3 - Aircraft Tracking

4.1 Objectives

- Set up Pi 4 as dedicated ADS-B receiver
- Deploy readsb and tar1090
- Integrate with unified dashboard

4.2 Hardware Required

Item	Qty	Est. Cost	Source
Raspberry Pi 4 4GB	1	On hand	-
RTL-SDR Blog V4	1	$40	RTL-SDR.com
1090 MHz filter	1	$20	RTL-SDR.com
1090 MHz LNA	1	$25	RTL-SDR.com
ADS-B antenna	1	$30	Amazon

Phase 3 Estimated Cost: ~$125

4.3 Implementation Steps

☐ Flash DietPi to SD card

☐ Configure hostname: adsb-station, IP: 192.168.1.52

☐ Install readsb and tar1090

☐ Optimize gain settings

☐ Add tar1090 link to dashboard

5. Phase 4: Sensor Mesh Network

PHASE 4 - Distributed Sensing

5.1 Objectives

- Configure Main Controller as WiFi AP (10.0.0.0/24)
- Deploy MQTT broker + InfluxDB + Grafana
- Create ESP32 and Zero 2W sensor nodes
- Integrate rtl_433 for ISM band monitoring

5.2 Hardware Required

Item	Qty	Est. Cost	Source
ESP32-WROOM DevKit	2	On hand	-
Pi Zero 2W	2	On hand	-
USB WiFi adapters	2	$20	Amazon
Battery packs	3	$45	Amazon
Enclosures	3	$30	Amazon

Phase 4 Estimated Cost: ~$95

5.3 Implementation Steps

☐ Configure hostapd and dnsmasq on Main Controller

☐ Install Mosquitto, InfluxDB, Grafana

☐ Flash ESP32 with WiFi probe scanner

☐ Configure Zero 2W with Kismet remote capture

☐ Install and configure rtl_433

☐ Build Grafana dashboards

6. Phase 5: HaLow Mesh + Advanced

PHASE 5 - Extended Range and Resilience

6.1 Objectives

- Deploy HaLowLink as mesh gateway (915 MHz)
- Configure 2.4 GHz guest AP (isolated)
- Set up home base for WAN backup
- Implement dual-WAN failover (Starlink + HaLow)
- Build unified dashboard
- Test GridDown operation

6.2 Hardware Required

Item	Qty	Est. Cost	Source
Morse Micro HaLowLink	3	On hand	-
915 MHz antennas	3	$45	Various
Starlink	1	Variable	Starlink
Enclosure (home)	1	$40	Amazon

Phase 5 Estimated Cost: ~$115 + Starlink

6.3 Implementation Steps

☐ Flash OpenWRT to HaLowLink devices
☐ Configure vehicle HaLowLink (eth0, wlan0, halow0)
☐ Set up guest network isolation
☐ Configure home base HaLowLink
☐ Test mesh connectivity and range
☐ Implement WAN failover script
☐ Build unified web dashboard
☐ Test GridDown operation

7. Budget Summary

Phase	New Purchases	Cumulative
Phase 1: Core Vehicle System	$313	$313
Phase 2: KrakenSDR Integration	$520	$833
Phase 3: ADS-B Station	$125	$958
Phase 4: Sensor Mesh	$95	$1,053
Phase 5: HaLow Mesh	$115	$1,168
TOTAL	**$1,168**	**(excl. Starlink)**

Note: Estimated $1,500-2,000 worth of equipment already on hand.

8. Project Timeline

8.1 Key Milestones

Week	Milestone	Deliverable
3	Core system operational in vehicle	Scanner UI + Kismet
5	Direction finding capability online	Scanner → Kraken DF
6	Aircraft tracking operational	tar1090 dashboard
9	Distributed sensor mesh functional	MQTT + Grafana
12	Full system integration complete	Unified dashboard

9. Quick Reference

9.1 IP Addresses

- Starlink Router: 192.168.1.1 (Gateway)
- Main Controller: 192.168.1.50 (+ 10.0.0.1 for sensors)
- Kraken Host: 192.168.1.51
- ADS-B Station: 192.168.1.52
- HaLowLink (Vehicle): 192.168.1.53, 192.168.2.1, 172.16.0.1
- HaLowLink (Home): 172.16.0.254

9.2 Service URLs

- Main Dashboard: http://main-controller.local/
- Scanner UI: http://main-controller.local:8080/
- Kismet: http://main-controller.local:2501/
- Grafana: http://main-controller.local:3000/
- Kraken DOA: http://kraken-host.local:8080/
- tar1090: http://adsb-station.local/tar1090/

9.3 Order Early (Long Lead Time)

- KrakenSDR (CrowdSupply) - 2-4 weeks
- 1090 MHz filter and LNA
- Quality 12V→5V converters
- Pelican case